Namani Vamshi Krishna

Software Engineer

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Summary

Providing comprehensive technical support to research scholars, specialized in writing research articles, support of publications, and development of code using Python, Verilog, and Matlab.

Work Experience

Software Engineer Oct 2020
Sak Informatics, Hyderabad, India Present

- Delivered innovative technical solutions across AI, IoT, and VLSI domains, contributing to advanced research and development projects.
- Developed and optimized Python-based algorithms for real-time data processing, analysis, and model evaluation.
- Utilized key Python libraries including NumPy, Pandas, and Matplotlib to process, analyze, and visualize complex datasets.
- Collaborated with cross-functional teams to integrate AI-driven insights into IoT and VLSI systems, enhancing performance and scalability.
- Conducted performance benchmarking and implemented optimization techniques to improve computational efficiency and model accuracy.
- Designed and fine-tuned Transformer-based deep learning architectures using PyTorch for high-performance feature extraction and classification tasks.
- Implemented advanced optimization workflows using the Mealpy metaheuristic optimization library to enhance model convergence and stability.
- Developed end-to-end pipelines for Transformer-based NLP and signal processing applications, leveraging pre-trained models and fine-tuning for domain-specific datasets.
- Applied iModels for interpretable machine learning, combining transparency with performance to generate explainable AI insights for model validation and decision support.

Trainee Jun 2015
Maven Silicon Jun 2016

- Gained hands-on experience in VLSI design and verification, developing and testing digital circuits and system architectures.
- Created Python scripts and automation tools to streamline verification workflows, data analysis, and simulation result processing.
- Collaborated with cross-functional teams to identify design issues, perform debugging, and optimize circuit performance.

- Applied standard verification methodologies to ensure accuracy, reliability, and compliance with design specifications.
- Utilized tools and frameworks to analyze large datasets, generate reports, and improve the efficiency of design verification processes.

Education

Ph.D. in Artificial Intelligence Oct 2022
Nirwan University, Present
Master of Technology (M.Tech) Jun 2016
Vignana Bharathi Institute of Technology, Oct 2018, A+

Skills

Python Skills

- TensorFlow, PyTorch, Keras, JAX, Theano
- Scikit-learn, XGBoost, LightGBM, CatBoost, Optun, NumPy, Pandas, SciPy, Statsmodels, SymPy
- Matplotlib, Seaborn, Plotly, Altair, Bokeh, Librosa, Torchaudio, Transformers, imodels, imbalanced-learn, flask, django, NGboost, keras, h5py, ARIMA

AI & Data Intelligence

- Transformer-based Deep Learning Models (BERT, ViT, Swin Transformer, GPT)
- MNLP, Audio and Speech Signal Analysis
- Sentiment Analysis using Transformers
- Time-Series Forecasting with Temporal Transformers

Embedded & IoT Systems

- IoT Data Management
- Real-Time Sensor Data Processing
- Edge AI for IoT Applications

Software & Development Tools

- Xilinx Vivado Design Suite, Matlab, Cadence, System VUE
- VLSI Back End Tools

Full Stack Development

- Flask, Django
- MySQL, Redis, LMDB

Publications

- **Nanotechnology-Enhanced Multipliers: Leveraging FinFET and GnrFET in Advanced Digital Systems**, IEEE, Nov 2024
- **Design of Hybrid Memory Built-in Self Test using Linear Feedback Shift Registers**, IEEE, Dec 2022
- **Medical Image Contrast Enhancement using Tuned Fuzzy Logic Intensification for COVID-19 Detection Applications**, IEEE, Feb 2022
- **SLDCNet: Skin Lesion Detection and Classification using Full Resolution Convolutional Network-based Deep Learning CNN with Transfer Learning**, Wiley, Jun 2022, Link

- **Zadoff-Chu Sequence Based Preamble Design of FBMC/OQAM for 5G**, International Journal of Engineering and Technology (UAE), Jan 2018, Link

ORCID and Scopus

- ORCID: 0000-0001-7872-5667
- Scopus: 57427494100

Workshops

- AI and ML-based CART Models Building with Real-Time Data, CMREC, 2024
- Real-Time Applications towards Transformers, MREM CSE-AIML, 2025
- Emerging Technologies in VLSI with AI Applications, SMEC ECE, 2024
- NLP-Based Sentiment Analysis with Multivariate Data, KPRIT CSE, 2023
- Business Analytics with Data Visualizations and XAI Analysis, KPRIT DS, 2024
- Transfer Learning-Based Computer Vision for Robotics and Medical Applications, MREM ECE, 2023
- Biomedical, Speech, and Signal Processing with AI, MRECW ECE, 2022
- IoT with Embedded Systems and Reinforcement Learning, SMEC ECE, 2022

Projects

- Smart Traffic Violation Detection using Vision Transformers
- Real-Time Weather Forecasting using Temporal Fusion Transformer
- Adaptive Voice Pathology Detection using Audio Spectrogram Transformer
- Intelligent Healthcare Monitoring using Swin Transformer
- Sentiment-Driven Chatbot using BERT Transformer
- Transformer-based Anomaly Detection in Industrial IoT Systems
- Sensor Inspection in Semiconductor Fabrication using DeiT
- Explainable Business Forecasting using GPT-Time Series Transformer